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Dr. Jia Min Chin
Department of Functional Materials and Catalysis
Faculty of Chemistry, Universität Wien
Vienna, Austria

Re: University Assistant Predoctoral — PhD Position in Inorganic & Materials Chemistry (Job-ID 5571)

Dear Dr. Chin,

I am applying for the predoctoral position in your group. My background spans colloidal self-assembly, protein and peptide handling, mass spectrometry characterisation, and a first-principles framework for predicting the spectroscopic properties of novel molecular structures without reference standards — a combination I believe connects directly to the specific challenges of field-driven MOF/COF synthesis.

Colloidal synthesis and self-assembly. During my MSc in Pharmaceutical Biotechnology at HAW Hamburg, I worked on the encapsulation of proteins inside inverse micelles: selecting the surfactant system, controlling the water-to-surfactant ratio w_0 , and tuning the hydrophilic core environment to stabilise a folded protein in a non-polar medium. This is supramolecular self-assembly in the same physical sense as MOF/COF growth — a cavity with defined chemical geometry nucleates around a guest molecule or forms through cooperative non-covalent interactions among building blocks. The specific challenge of controlling cavity size, polarity, and guest compatibility in inverse micelles is structurally identical to the pore engineering problem in framework materials, and the experience of manipulating assembly conditions — solvent composition, concentration, temperature, and field effects — translates directly. I have also prepared and characterised lipid nanoparticles and liposomes (DLS, zeta potential, electron microscopy) and worked on the production and characterisation of therapeutic lipid colloids at Fraunhofer Institut IGB. The colloidal assembly dimension of your project is therefore familiar ground.

DNA and peptide handling. The position lists DNA or peptide handling as a preferred background. Across several roles I have worked directly with proteins and peptides: at the University Medical Center Groningen I conducted UPLC-MS/MS shotgun proteomics, which requires enzymatic protein digestion, peptide fractionation, and LC separation of complex peptide mixtures; at Universitätsklinikum Hamburg-Eppendorf I built automated characterisation pipelines for proteomics and glycomics samples from LC-MS/MS and MALDI-TOF data, again working with digested proteins and peptide-level identification. The inverse micelle project at HAW Hamburg involved stabilising folded proteins inside the micelle core — handling proteins under conditions designed to preserve their native structure in a non-aqueous environment. These roles establish direct hands-on experi-

ence with peptides and proteins as the relevant material class for DNA/peptide-directed MOF/COF synthesis.

Spectroscopic characterisation of novel framework structures. When a new MOF/COF linker combination is synthesised, no reference IR or Raman spectrum exists yet in any library. The Borgia framework I have developed addresses this directly: it derives molecular vibrational modes and harmonic coupling from bounded phase space geometry alone, requiring no empirical calibration against reference data. The foundational result is that molecular absorption-emission intervals create intrinsic frequency ticks; modes sharing rational frequency ratios $\omega_i/\omega_j = p/q$ ($\max(p, q) \leq 10$) form coupled harmonic networks with closed loops that act as virtual resonant cavities. This has been validated on six molecules (H_2 through C_6H_6) against NIST spectroscopic data with self-consistency errors below 1.63% and zero adjustable parameters. For a novel MOF linker, the framework predicts which vibrational modes will harmonically couple, providing theoretical fingerprints for IR/Raman assignment and identifying which coupling patterns are diagnostic for specific coordination motifs in the grown framework. The cheminformatics instrument built on this framework is available in the browser: zero-parameter molecular spectroscopy including vibrational modes, docking, 3D structure, and harmonic network analysis, requiring no installation or account. A companion paper — *Harmonic-Scattering Loop Coupling* — extends this to a full transfer-matrix framework for coupled molecular resonators, achieving machine-precision reconstruction across 18/18 benchmarks; and a Resonant Stack instrument demonstrates the physical consequence: Beer–Lambert absorption, chromatographic separation, and Kirchhoff radiation law are mathematically identical operations in partition coordinates. For a novel MOF/COF linker this means the vibrational mode predictions cover IR absorption, Raman scattering, and thermal emission simultaneously — the same first-principles calculation, no re-calibration per spectroscopic method.

Field-driven alignment and charge transport anisotropy. Your project specifies three field types: electric, magnetic, and flow. The flow case is addressed directly by the Gas Giant platform I have built, which derives fluid dynamics from partition network interference — Kirchhoff circuit laws map to Navier-Stokes; viscosity emerges as $\mu = \tau_c \times g$ without phenomenological fitting; Poiseuille flow profiles validate to 0.9% mean error. For flow-directed MOF/COF assembly, the flow field imposes a preferred partition direction through viscous alignment forces, and the partition lag τ_c — the same parameter that governs viscosity — determines the alignment timescale as a function of flow velocity and particle size. The physical mechanism by which electric and magnetic fields direct growth is that the field breaks rotational symmetry of the assembly environment, biasing crystal growth along a preferred axis. In the partition coordinate framework underlying Borgia, this is equivalent to imposing a preferred partition direction: the coordination environment along the field axis develops a deeper, more nested structure than the transverse directions. Charge transport anisotropy follows because conductivity along the field axis reflects the higher partition depth of that structural direction. The composition-inflation theorem ($T(n, d) = d(d+1)^{n-1}$) predicts that the anisotropy ratio scales combinatorially with the number of field-aligned assembly cycles — a quantitative, testable connection between synthesis parameters (field strength, duration) and the resulting transport tensor that could be validated against the electrochemical sensing data your group generates.

Background. I hold an MSc in Pharmaceutical Biotechnology (HAW Hamburg), an MSc in Computational Biology (Universität Tübingen), and completed doctoral research in

Computational Lipidomics at TUM. I am legally authorised to work in Austria and am available from June 2026. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye